Study Project on

Simpson's Integration Approach for Geometric Nonlinear Sensitivity

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Contents

- Introduction
- Nonlinear Strain Energy
- Simpson-Based Formulation
- Results
- Inferences

Contents

- **Introduction**
  - Structural Optimization
  - Geometric Linear Objective
  - Geometric Non-Linear Objective

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Introduction

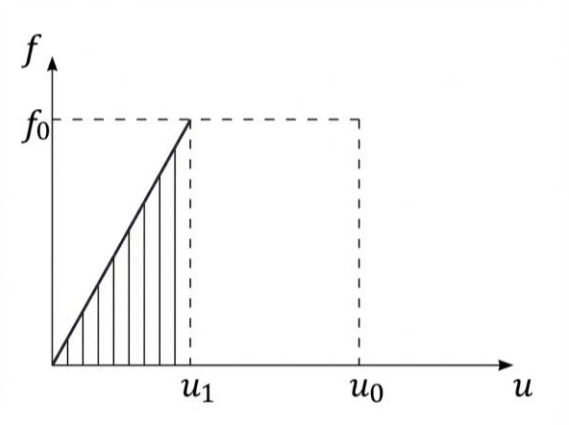
Structural Optimization

- Objective: Increase stiffness → Minimise strain energy
- Geometric Linear
- Geometric Non-Linear

Introduction

Geometric Linear Objective

- Objective function: $\frac{1}{2}$ KU^2
- Linear Sensitivity Formulation: $\frac{1}{2}\ \mathrm{U}^{\mathrm{T}} \frac{\partial \mathrm{K}}{\partial s} \mathrm{U}$
- U – Displacement Vector, S - Design Variable
- Optimized Structures may be prone to buckling

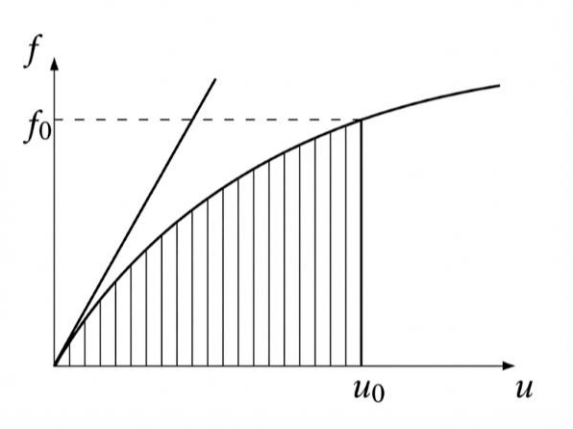


Introduction

Geometric Non-Linear Objective

- Optimization considering geometric nonlinear behaviour
- Flexible structures
- Objective formulations - **Exact Nonlinear**

  - End compliance

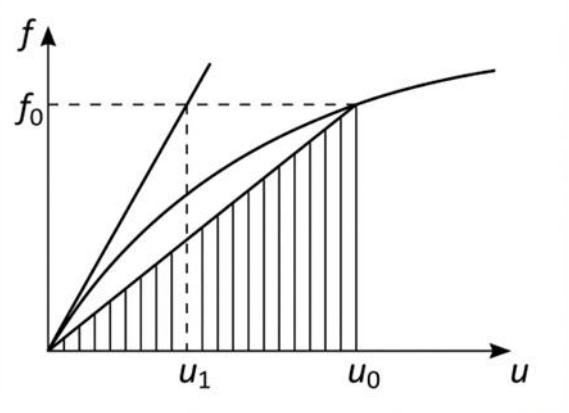


Introduction

Geometric Non-Linear Objective

- Optimization considering geometric nonlinear behaviour
- Flexible structures
- Objective formulations - Exact Nonlinear

  - **End compliance**
- Load-displacement history is neglected
- Structure Optimized for the specific selected load
- Results in adverse optimized shape for structures dominated by snap-through buckling



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- **Nonlinear Strain Energy**
  - Trapezoidal approach
  - Proposal - Simpsons integration approach

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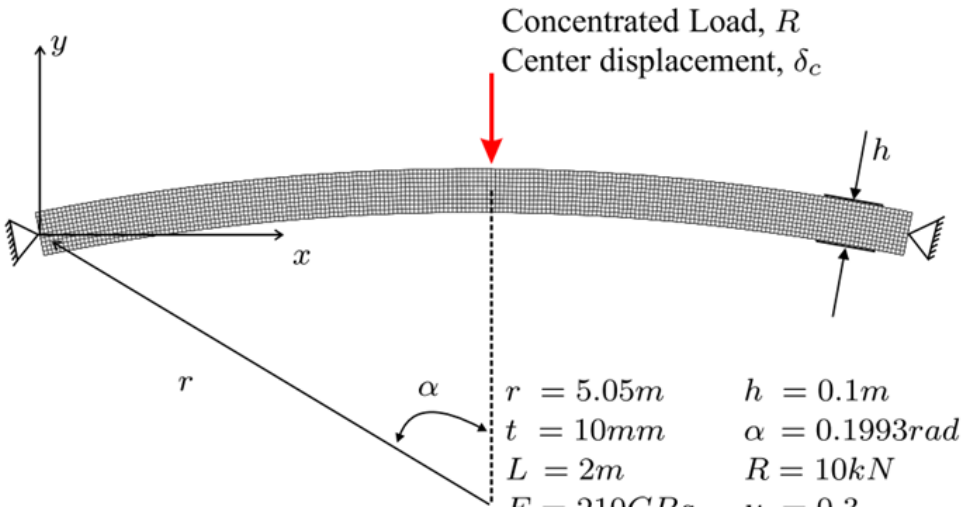
Nonlinear Strain Energy

Motivation - Load dependency



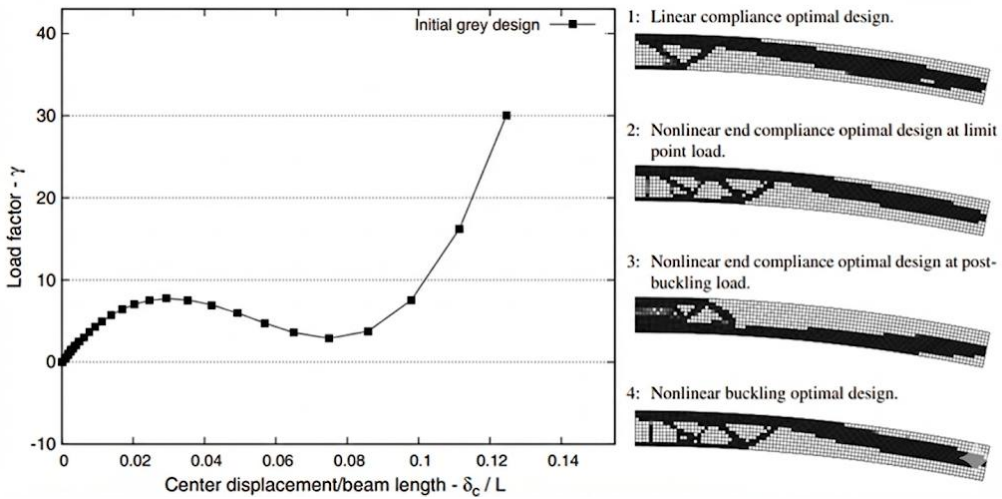
Load deflection Curve [Reference – 1, Fig.2]

Study snap-through behavior with different Objective formulations

- Optimization- 1: Linear compliance
- Optimization- 2: Nonlinear end compliance at limit load
- Optimization- 3: Nonlinear end compliance at post-buckling load
- Optimization- 4: Nonlinear Buckling

Nonlinear Strain Energy

Motivation - Problem description



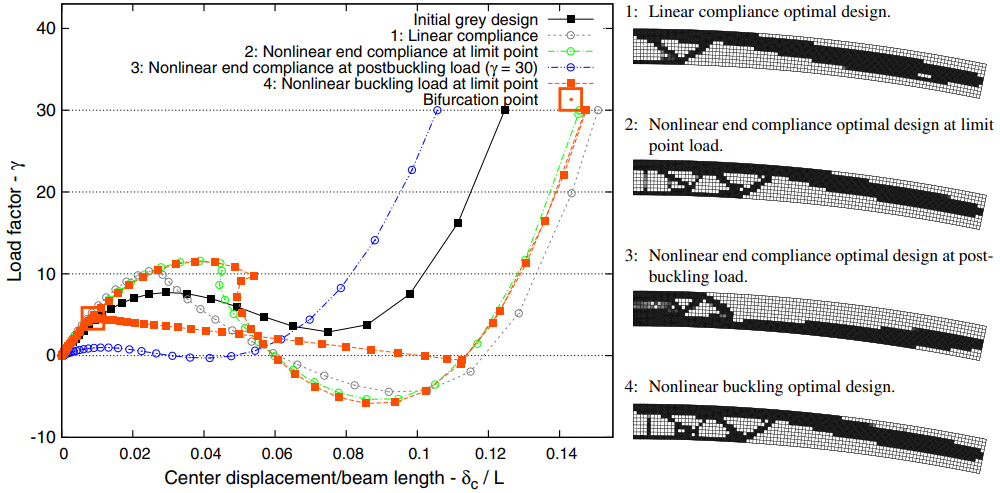
Load deflection Curve [Reference – 1, Fig.3]

Study snap-through behavior with different Objective formulations

- Optimization- 1: Linear compliance
- Optimization- 2: Nonlinear end compliance at **limit load**
- Optimization- 3: Nonlinear end compliance at **post-buckling load**
- Optimization- 4: Nonlinear Buckling

Nonlinear Strain Energy

Motivation - Load dependency

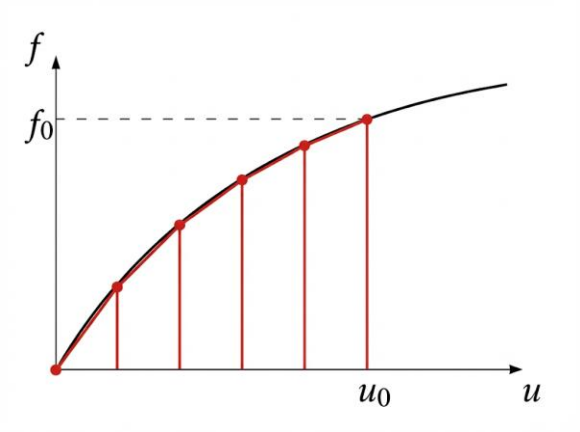


Load deflection Curve [Reference – 1, Fig.3]

- Topology Optimized shape for Optimization - 2 Vs Optimization - 3
- Load Displacement Curve for Optimization - 2 Vs Optimization - 3
- Optimization - 2: Higher buckling load & Lower post-buckling Strength
- Optimization - 3: Lower buckling load & Higher post-buckling Strength

Nonlinear Strain Energy

Trapezoidal approach

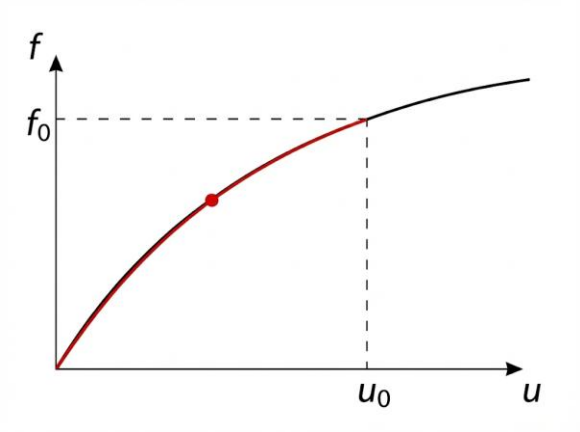


- Advantage: Excat Nonlinear representation of structure in the objective function

- Drawback: Time-consuming - number of computations of displacement per optimization iteration increases

Nonlinear Strain Energy

Proposal - Simpsons integration approach



Area = $\frac{1}{6} * x * (4y_1 + y_2)$

- Advantage: 1. Excat up to 3rd degree polynomial

  2. Nonlinear computations per iteration is reduced

- Drawback: 1. Inaccurate for degree greater than 3

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- **Simpson-Based Formulation**
  - Methodology
  - Objective Formulation
  - Adjoint Sensitivity Formulation
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Simpson-Based Formulation

Methodology

- Objective function formulation for Simpson’s approach
- Analytical adjoint sensitivity derivation
- Comparison of Finite Difference (FD) sensitivity of Simpson’s with the Trapezoidal approach
- Verify analytical sensitivity of Simpson’s by comparing against its FD values

Simpson-Based Formulation

Objective Formulation

Area under a quadratic curve by Simpsons = $\frac{1}{6}$ * x * ($4y_1 + y_2$), assuming start point (0,0)

Condition: y_1 must be found at $X_1 = \frac{X_2}{2}$.

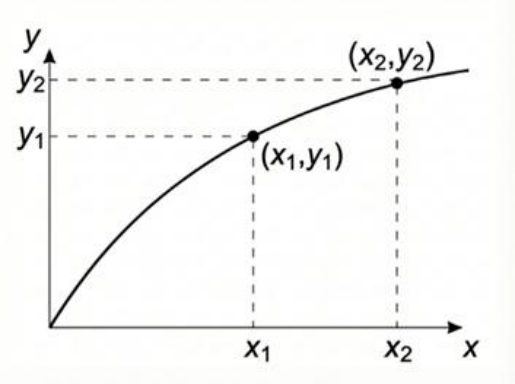
For our problem, y is load, F

x is displacement, u

But here, x is taken as the load(F)

y is taken as the displacement(u)

Complementary Strain Energy



Reason: Implemented software capability - Load control only

Potential Issue: Find **[F]** which satisfies **[U_1] =** $\frac{[U_2]}{2}$ for all u values and **[F_n]= λ[F]**

Simpson-Based Formulation

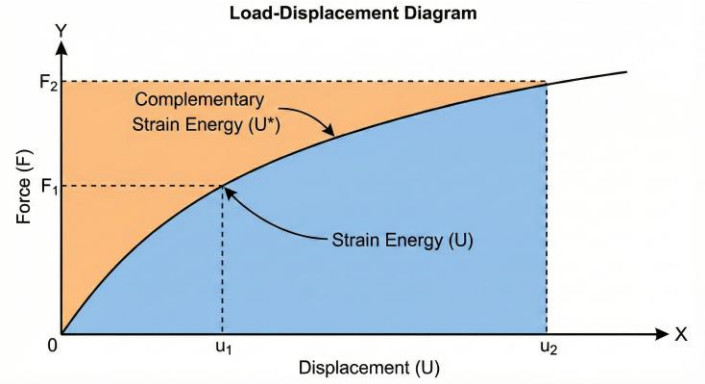
Objective Formulation

Then,

$$\text{Area} = \frac{1}{6} * F_2 * (4U_1 + U_2) \longrightarrow \text{Complimentary Strain Energy}$$

U_2 is the nonlinear displacement for force F_2

U_1 is the nonlinear displacement for force $F_1 = \frac{F_2}{2}$.



$$\text{Strain Energy} = F_2 U_2 - \frac{1}{6} * F_2 * (4U_1 + U_2)$$

$$= [F_2]^T[U_2] - \frac{1}{6} * [F_2]^T * (4[U_1] + [U_2])$$

Simpson-Based Formulation

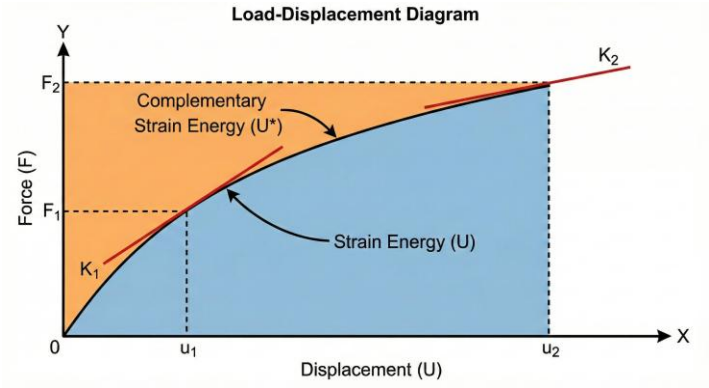
Adjoint Sensitivity Formulation

Then,

Complimentary $\frac{dJ}{dS} = \left[\frac{4}{6} * F_2^T * K_1^{-1} * \frac{dK_1}{dS} * U_1\right] + \left[\frac{1}{6} * F_2^T * K_2^{-1} * \frac{dK_2}{dS} * U_2\right]$

Adjoint analytical sensitivity equation,

Complimentary $\frac{dJ}{dS} = \left[\frac{4}{6} * (K_1^{-1}F_2)^T * \frac{dK_1}{dS} * U_1\right] + \left[\frac{1}{6} * (K_2^{-1}F_2)^T * \frac{dK_2}{dS} * U_2\right]$



$$\frac{dJ}{dS} = \left[U_2^T * \frac{dK_2}{dS} * U_2\right] - \left[\left[\frac{4}{6} * (K_1^{-1}F_2)^T * \frac{dK_1}{dS} * U_1\right] + \left[\frac{1}{6} * (K_2^{-1}F_2)^T * \frac{dK_2}{dS} * U_2\right]\right]$$

K_1^{-1} and K_2^{-1} - Two inverse computations

Simpson-Based Formulation

Adjoint Sensitivity Formulation

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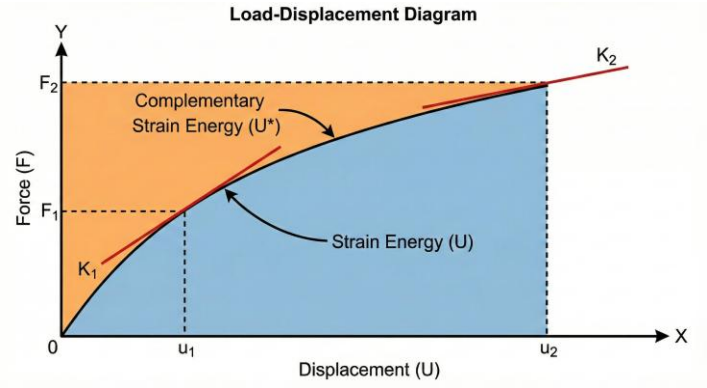


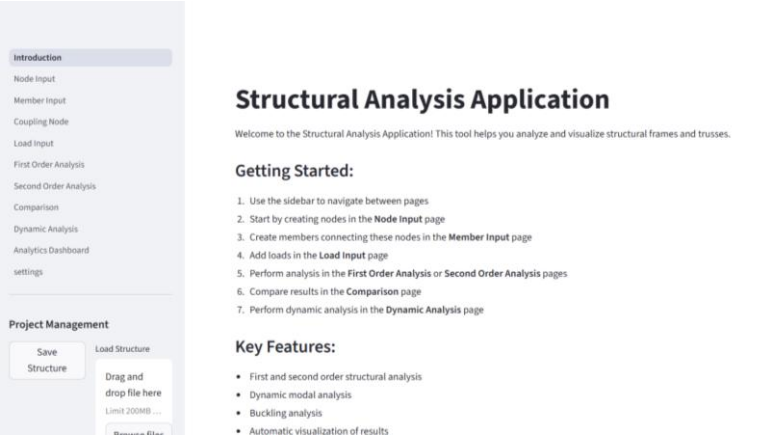
$$\boldsymbol{\frac{dJ}{dS} = \left[U_2^T * \frac{dK_2}{dS} * U_2\right] - \left[\left[\frac{4}{6} * (K_1^{-1}F_2)^T * \frac{dK_1}{dS} * U_1\right] + \left[\frac{1}{6} * (U_2)^T * \frac{dK_2}{dS} * U_2\right]\right]}$$

K_1^{-1} and K_2^{-1} - **Two inverse** computations

Simpson-Based Formulation

Implementation

- Implemented in self-developed software
- Capabilities
  1. 2D - 1st and 2nd order Structural Analysis
  2. Dynamic Modal Analysis
  3. Linearized Buckling analysis
- Available as
  1. Website
  2. PyPi module
- Programming language – Python
- Element – 2nd order Bernoulli beam element



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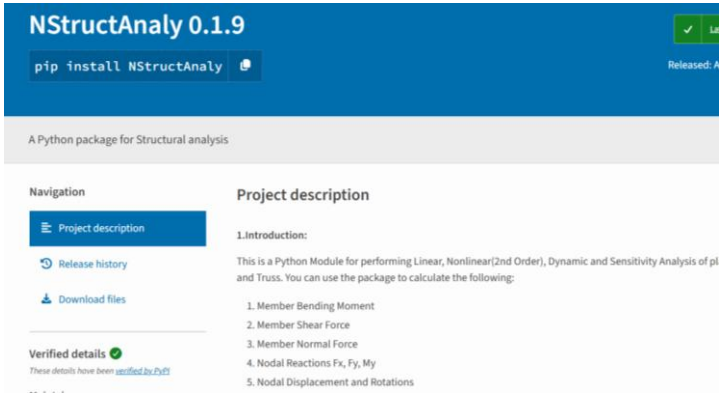

Simpson-Based Formulation

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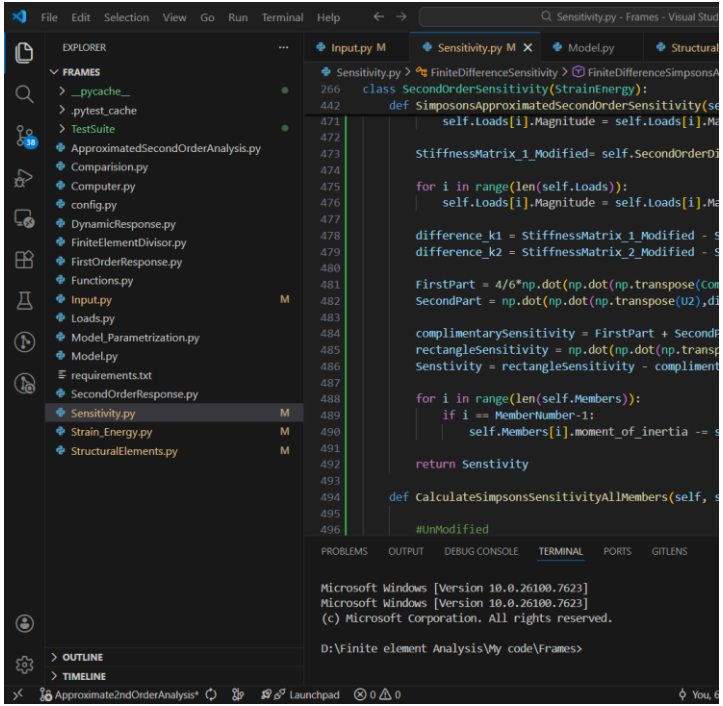

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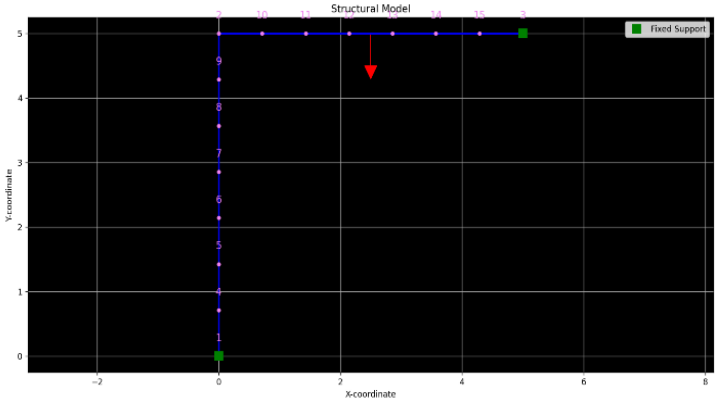
  - FD Sensitivity - Linearized End Compliance
  - Simpsons - Adjoint vs FD Sensitivity
- Inferences

Results

Models

Model 1



Support condition: Both Fixed

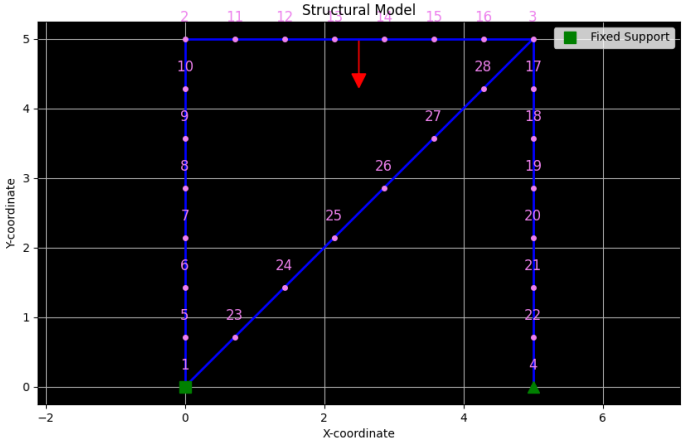
Design variable : Moment of inertia

Discretization : 7 elements per beam

Area : 0.09

Moment of inertia: 0.000675

Model 2



Support condition: Fixed and Hinged

Design variable : Moment of inertia

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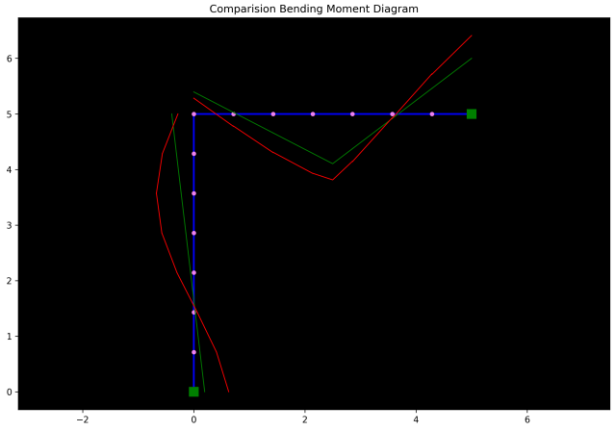
Results

BMD

Model 1



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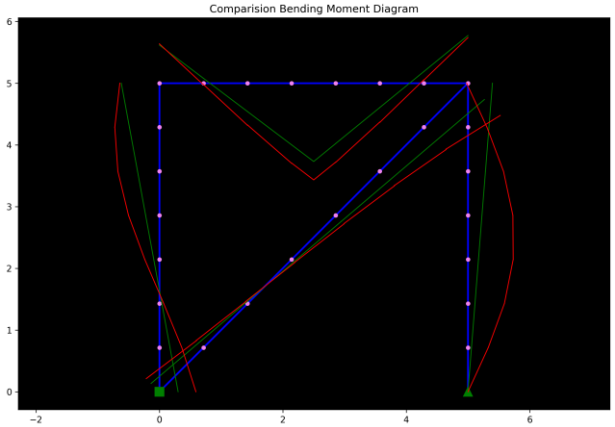
Model 2



Support condition: Fixed and Hinged

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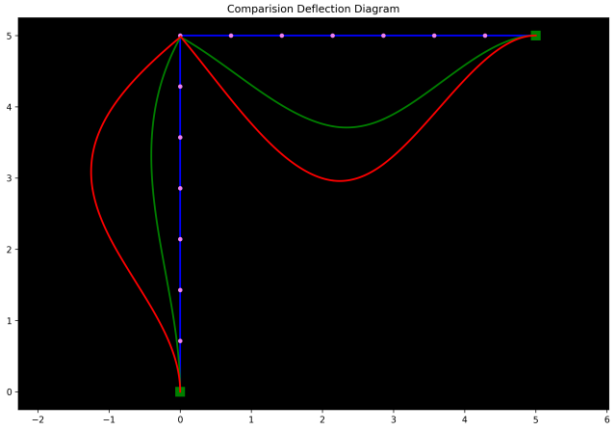
Results

Deflection

Model 1



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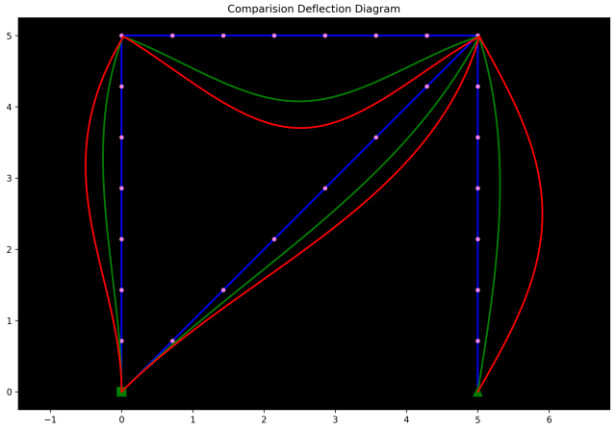
Model 2



Support condition: Fixed and Hinged

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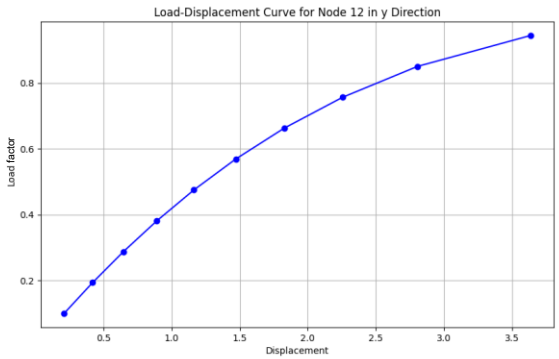
Results

Load Displacement Curve

Model 1



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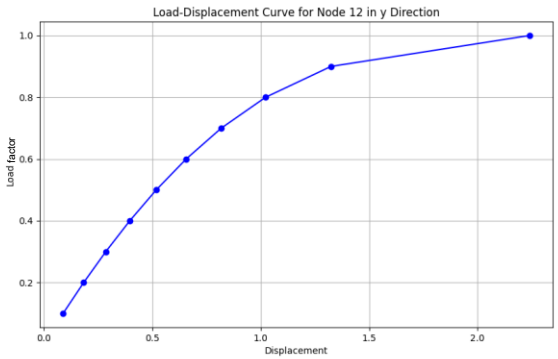
Model 2



Support condition: Fixed and Hinged

Design variable : Moment of inertia

Discretization : 7 elements per beam

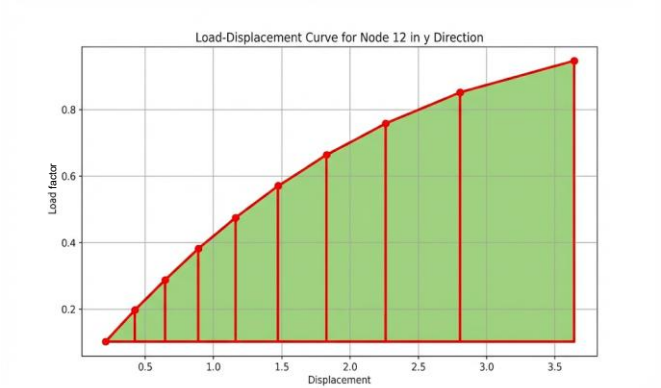
Area : 0.09

Moment of inertia: 0.000675

Results

FD Sensitivity - Trapezoidal Approach - Reference

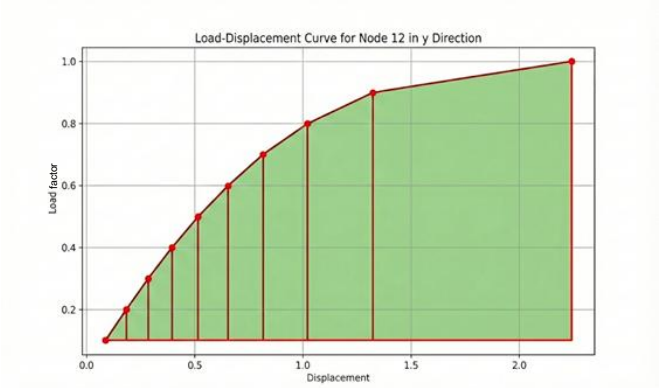
Model 1



Objective: Green region

Divisions: 10

Model 2



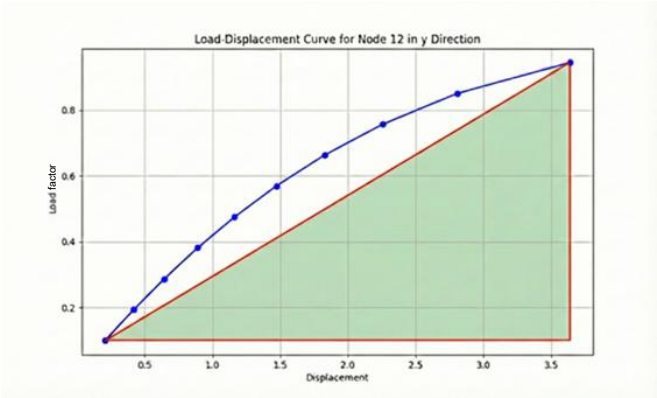
Objective: Green region

Divisions: 10

Results

FD Sensitivity - Linearized End Compliance

Model 1

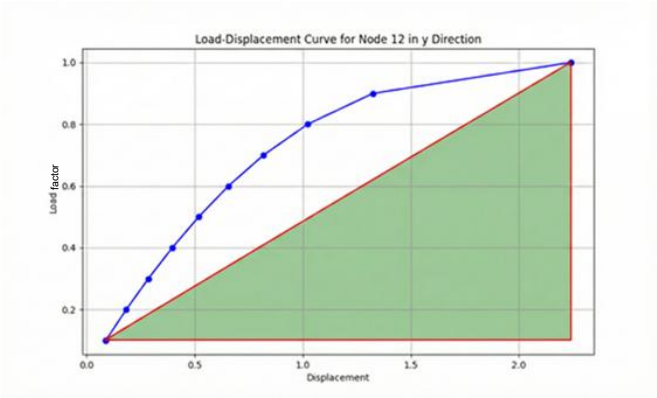


Objective: Green region

Divisions: 1

Average Error in sensitivity: 43.11 %

Model 2



Objective: Green region

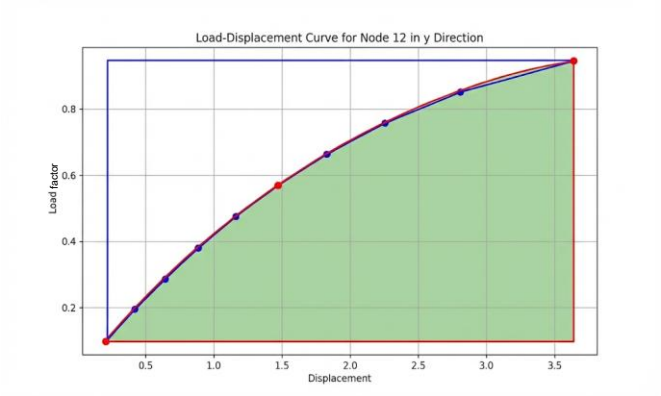
Divisions: 1

Average Error in sensitivity: 54.96 %

Results

FD Sensitivity - Simpson's Approach

Model 1

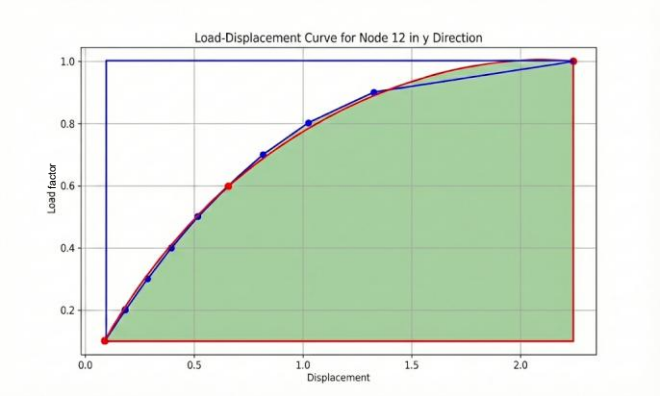


Objective: Green region

Divisions: 2

Average Error in sensitivity: 9.78 %

Model 2



Objective: Green region

Divisions: 2

Average Error in sensitivity: 9.03 %

Results

Simpsons - Adjoint vs FD Sensitivity

Element	Analytical Simpsons	Numerical Simpsons	Relative Error (%)
1	-6.85E-05	-6.85E-05	0
2	-1.64E-05	-1.64E-05	0.03
3	-6.44E-06	-6.44E-06	0.04
4	-5.05E-05	-5.05E-05	0.01
5	-9.71E-05	-9.71E-05	0.01
6	-9.08E-05	-9.08E-05	0
7	-3.68E-05	-3.68E-05	0.01
8	-2.16E-06	-2.16E-06	0.08
9	-5.53E-05	-5.53E-05	0
10	-1.75E-04	-1.75E-04	0.01
11	-2.40E-04	-2.40E-04	0
12	-5.42E-05	-5.42E-05	0.01
13	-3.38E-05	-3.38E-05	0
14	-2.47E-04	-2.47E-04	0

$$\frac{dJ}{dS} = \left[U_2^T * \frac{dK_2}{dS} * U_2\right] - \left[\left[\frac{4}{6} * (K_1^{-1}F_2)^T * \frac{dK_1}{dS} * U_1\right] + \left[\frac{1}{6} * (U_2)^T * \frac{dK_2}{dS} * U_2\right]\right]$$

- Model - 1 Avg error - 0.01%
- Model - 2 Avg error - 0.08%

Derived Adjoint Sensitivity is Valid

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Inference

1. Simpson's approach is efficient than the trapezoidal approach.

2. With twice the increase in computational cost, the error can be decreased up to 50 percent.

3. Complimentary strain energy-based objective formulation - Not applicable to snap-through problems

Thank You